

What Makes a Meme Relieving?: Analyzing the Efficacy of Design Elements of Internet Memes on Exam Stress Relief for University Students

Zuyu Xu

Hong Kong University of Science and Technology
Hong Kong, China
zxubo@connect.ust.hk

Shixu Zhou

Hong Kong University of Science and Technology
Hong Kong, China
szhouav@connect.ust.hk

Jiaxiong Hu

Hong Kong University of Science and Technology
Hong Kong, China
hujx@ust.hk

Xiaojuan Ma*

Hong Kong University of Science and Technology
Hong Kong, Hong Kong
mxj@cse.ust.hk

Abstract

Exam stress is a significant issue for university students, often influencing academic performance and well-being. During preparation for test-taking situations, students may prefer support that is quick and low-effort. To better understand how internet memes might serve as lightweight, complementary support in this context, we studied exam-related image memes and the design elements associated with perceived stress-relief effectiveness. We collected a multi-platform dataset of 252 exam-related memes and developed an annotation codebook through iterative qualitative content analysis, identifying ten categories of meme design elements. We then conducted a crowdsourcing study in which students rated memes' perceived effectiveness for reducing exam stress, and used feature analysis to examine how annotated design elements relate to effectiveness labels. Our results identify a set of informative design elements associated with higher perceived stress-relief effectiveness and provide actionable insights for meme content design and curation in exam-preparation contexts.

CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design**; *Interactive systems and tools*.

Keywords

Exam stress, Digital micro-intervention, University students, Internet memes

ACM Reference Format:

Zuyu Xu, Jiaxiong Hu, Shixu Zhou, and Xiaojuan Ma. 2026. What Makes a Meme Relieving?: Analyzing the Efficacy of Design Elements of Internet Memes on Exam Stress Relief for University Students. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798612>

*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798612>

1 Introduction

Exam stress refers to the physiological and behavioral responses that accompany concern about potential negative outcomes under evaluative circumstances, including examinations, project presentations, paper submission, etc. [25, 26, 30]. It is common among university students [17], and can hinder academic performance while undermining self-esteem and confidence [26, 30]. Despite broad recognition of these impacts, the period leading up to exams and deadlines poses practical constraints for support. Students during exam preparation are often time-constrained, cognitively loaded, and reluctant to engage in interventions that are complex or time-consuming, as they prefer to focus on preparation [25]. Also, prior work on exam-related stress and test anxiety has examined a range of stress-relief strategies, including coping skills and cognitive approaches that aim to reduce maladaptive appraisals and improve emotion regulation [9, 10, 18, 26, 30]. However, many such strategies require sustained practice, structured guidance, or timely access to support, which can be difficult to maintain during intensive exam preparation [4, 25]. These limitations create a need for stress-support approaches that are lightweight, quick to engage with, and compatible with fragmented study routines.

Internet memes—highly recognizable and humor-oriented image-text artifacts—are increasingly used in everyday online communication to express feelings and shared experiences [7, 12, 28]. For university students, academic-life memes are prevalent online and require minimal time to engage with, suggesting their potential as a lightweight, complementary resource during exam preparation [3, 23, 28]. With inherent humor and contextual relevance, Internet memes comfort readers' emotions, alleviate their tension, and distract them from interpersonal struggles [2, 20, 24]. In stressful contexts, memes may support coping-related experiences such as brief distraction and humorous reframing, and can foster perceived social connection through shared, relatable narratives [2, 20, 24]. From an emotion-regulation perspective, humorous reframing can function as a form of reappraisal, suggesting a plausible pathway through which meme viewing could be perceived as stress-supportive during exam preparation [10, 18]. Previous studies have increasingly recognized the psychological benefits of Internet memes for online members bearing mental health difficulties [1, 2, 24]. However, these studies rarely examined how specific

meme design elements shape students' perceived stress support. For one thing, not all memes may positively benefit mental health regulation. Dark or hateful memes, which feature abusive or offensive content, may lower the mood state and increase sensitivity towards anxiety [14]. In parallel, prior work has developed operational frameworks to systematically characterize image memes [19, 29]. For instance, Ling et al. [19] developed a codebook of visual indicators to systematically characterize image memes and distinguish highly viral from less viral content. While these studies demonstrate that meme design elements can be reliably specified and analyzed, they primarily target popularity or engagement outcomes of memes. Therefore, to inform principled design rather than relying on intuition, we need an interpretable way to characterize meme designs and examine which design elements are associated with higher perceived stress-relief potential in exam-related scenarios.

Building on this motivation, we investigate exam-related image memes through a mixed-method approach. We first curate a multi-platform dataset of 252 exam-related memes and develop an annotation codebook via iterative qualitative content analysis. We then conduct a crowdsourcing study to capture students' perceptions of memes' effectiveness for reducing exam stress in exam-preparation scenarios. Finally, we analyze how annotated design elements relate to perceived effectiveness and identify the most informative elements to support evidence-based meme design for lightweight, complementary support in evaluative contexts. Overall, our findings offer HCI researchers and practitioners a concrete vocabulary and initial evidence base for designing and selecting meme-based content to support students during exam preparation.

2 Design Elements of Memes for Exam Stress Reduction

In this section, we describe two studies that ground our analysis of exam-related image memes. We first curate a multi-platform dataset (Sec 2.1) and develop a meme design-element codebook through iterative qualitative content analysis (Sec 2.2). We then conduct a crowdsourcing study to capture students' perceived effectiveness of these memes for reducing exam stress in exam-preparation scenarios (Sec 2.3) Finally, we use feature analysis to examine how annotated design elements relate to perceived effectiveness (Sec 2.4).

2.1 Dataset Collection

To compile a representative dataset of exam-related memes, we adopted a multi-source collection strategy combining international, Chinese-language, and meme-specific platforms. International sources included *Reddit*, *Instagram*, *9GAG*, *Tumblr*, while Chinese-language platforms comprised *Weibo*, and *Xiaohongshu*. We also incorporated meme-dedicated sites including *memes.tw*, *Imgflip*, and *Imgur* to expand format diversity and include content specifically created for meme sharing. These sources were selected based on three criteria: (1) high popularity among university-aged users, (2) active circulation of humor related to academic life and exams, and (3) complementary coverage of Western and Asian cultural contexts.

We constructed search queries by pairing one keyword from an academic event category with one keyword from a stress-related category. The first type concerns common academic events with a deadline, including *Exam*, *Deadline*, *Presentation*, *Paper Submission*,

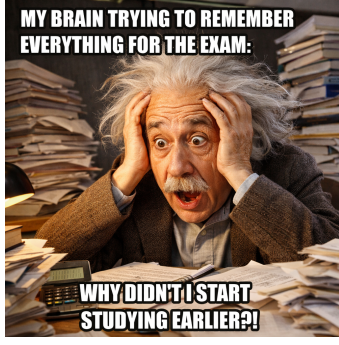
etc. The second type is keywords related to stress, including *Stress*, *Anxiety*, and *Depression*. All platforms were searched using these paired terms to ensure the retrieval of memes reflecting both exam-related situations and emotional stress. Searches were conducted using the built-in search tools or public APIs where available, with manual browsing employed when platform indexing was limited. We restricted results to content posted between January 2019 and December 2023 to capture contemporary meme conventions while keeping the corpus fixed and reproducible for screening, annotation, and evaluation. Also, we set the cutoff to reduce potential distribution shifts from the rapid uptake of generative AI tools in meme creation.

The initial search yielded 278 memes. Two independent reviewers then screened the dataset using predefined inclusion/exclusion criteria, including removing duplicates, disrespectful or discriminatory content, non-English/Chinese memes, and low-quality images with unreadable text. Following this process, we retained 252 unique memes. This multi-platform dataset provided a diverse and culturally varied basis for subsequent content analysis, capturing both widely circulated mainstream memes and more niche content formats.

2.2 Content Analysis: Process and Results

We followed the process model of the qualitative content analysis presented by Mayring [21], and extracted design elements of an Internet meme through iterative coding. More precisely, we developed the codebook as follows. Before starting, all three coders familiarized themselves with all the memes in the dataset. We first randomly selected 40 out of 252 memes to obtain the initial codes. Each coder separately identified meaning units in the memes that had a potential effect on stress regulation and assigned labels to them inductively. Next, we met regularly and discussed these initial codes to compare their extracted meaning units and explanations, address overlapping or conflicting labels, eliminate redundancy, and check whether all aspects related to the aim had been covered. The process continued until the codebook reached stability and further iterations would not yield additional refinement. Then, we picked another 40 random memes and repeated the whole process to ensure that the codebook comprehensively captures all relevant patterns. After these two rounds of coding with label addition and revision wherever necessary, the coder met and performed the categorization of codes together. Upon completion, we identified 10 categories of design elements that characterized image memes with potential effects on exam stress reduction. The elements include **Humor Type**, **Subject**, **Cultural Specifics**, **Stressor-Relatedness**, **Pronoun**, **Panel Number**, **Shot**, **Movement**, **Textual Valence**, and **Character Emotion**. We provide definitions for each element and all value labels in the supplementary material.

To finalize the coding process and assess inter-coder reliability, two of the three coders independently assigned labels to the remaining unlabeled 172 memes based on the codebook. Following the independent coding, the third coder calculated Cohen's kappa [22] for each element to evaluate the agreement between the two coders and determine the reliability of the coding scheme, as shown in Table 1. The inter-coder agreement was generally high. Among all 46 labels, 12 have almost perfect agreement (score above 0.8),



(a) Humor Type: Exaggeration; Subject: Human; Panel: Single Panel

me at a job interview claiming I can work under pressure	also me working under pressure
---	--------------------------------------



(b) Humor Type: Contrast; Subject: Animal; Panel: Multiple Panels

Figure 1: Representative exam-related memes illustrating different design elements. Labels under each subfigure show a subset of salient elements (e.g., humor type, subject, panel number); see Table 1 and the supplementary material for the full set of elements, value labels, and operational definitions.

13 have substantial agreement (score between 0.6 and 0.8), and 18 have moderate agreement (score between 0.4 and 0.6). Only 3 labels (“Simple Metaphor”, “Sarcasm”, and “Causal Movement”) fell below the threshold of moderate agreement. We discussed together and analyzed the reason. For “Simple Metaphor” and “Sarcasm”, labeling often requires cultural or background knowledge, which may not be uniformly shared by all coders. For “Causal Movement”, coders had different levels of understanding when determining whether a specific movement truly reflects a causal relationship, as this requires more abstract reasoning. After discussion, we reached an agreement on the interpretation and application of these labels, ensuring better alignment with definitions in the codebook.

2.3 Crowdsourced Labeling Study

With the approval of our institution’s IRB, we conducted crowdsourcing tasks on Credamo, an online platform designed to provide data solutions for research and business analytics. We restricted the crowd workers to be graduates or undergraduates aged 18-26. We also limited them to being proficient in both Chinese and English. Participants first report their basic demographic information, including age and gender. Next, three different memes from our

dataset were randomly assigned to each participant. To eliminate potential mutual influences among memes, we inserted a neutral image between each pair of images, forming a *meme-neutral-meme-neutral-meme* structure. Specifically, one participant needed to rate three meme images and two neutral images. These two neutral images were selected from the International Affective Picture System (IAPS) [16], both denoted with neutral emotional valence ratings and proven to cause low emotional arousal [27]. We designed five scenarios describing stressful situations during testing preparation, each inserted before one image. These scenarios are intended to help participants recall similar experiences, enabling them to better evaluate the usefulness of memes in reducing exam stress. After reading the scenario description, participants were asked to observe the image and rate it on a scale from 0 to 100 for its effectiveness in reducing exam stress [6]. Participants are not allowed to view the image for the second time to avoid the influence of subsequent images. We also set one quality control question after each image rating task, in which participants were asked to choose a specific number from 1 to 5. We utilized this type of question to determine whether participants answered the questions seriously.

We filtered out invalid tasks if the answers failed to pass the quality control questions. Each participant will receive 0.28 USD for one valid task. After removing invalid tasks, we sorted the evaluation for each meme in chronological order and retained only the first five ratings. For images with fewer than five valid responses, we repeated the aforementioned crowdsourcing steps until each image had five valid responses. In total, 470 participants were recruited (28.1% male, 71.1% female, 0.2% prefer not to say, age mean $\pm$ SD = 22.1 ± 1.8).

2.4 Feature Analysis

We examined which meme design elements were most effective in predicting whether a given meme would be perceived as effective or ineffective in reducing exam stress. We calculated the mean of five valid ratings for each meme. Those with a mean greater than the mean of overall ratings are labeled as “effective”, and those below are labeled as “ineffective”. We then selected five commonly used machine learning classifiers, including K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), Logistic Regression, Multi-layer Perceptron (MLP), and Random Forest for training. For each classifier, we randomly split these memes into a training set (80%) and a testing set (20%). We determined the hyper-parameters by 10-fold cross-validation on the training set and calculated the mean accuracy with its standard deviation, precision, recall, and F1-score, as shown in the Table 2. We observed that Logistic Regression achieved the best overall performance, with the highest mean accuracy, precision, recall, and F1-score. We then conducted a feature analysis of this classifier and discussed the top five features learned by this model.

- **Close Up Scale:** The Logistic Regression algorithm identified “Close Up Scale” as the most significant feature for classification. 72.7% of effective memes were characterized by the presence of a “facial expression,” compared to only 9.09% of ineffective memes exhibiting this feature. Close-up shots of characters can capture detailed facial expressions,

Table 1: Inter-annotators agreement for each feature. *** indicates almost perfect agreement (Cohen’s Kappa values score above 0.8), ** substantial agreement (Cohen’s Kappa value between 0.6 and 0.8), * moderate agreement (Cohen’s Kappa value between 0.4 and 0.6), and + slight or fair agreement (Cohen’s Kappa value lower than 0.4)

Element	Label	Agreement	Element	Label	Agreement
E1. Humor Type	Metaphor	0.583 *	E5. Pronoun	No Pronoun	0.960 ***
	- Simple Metaphor	0.126 +		First Person Pronoun	0.916 ***
	- Personification	0.622 **		Second Person Pronoun	0.906 ***
	- Labeling	0.631 **		Third Person Pronoun	1.000 ***
	Exaggeration	0.578 *		Mixed Pronoun	0.702 **
	Contrast	0.745 **		Dialogs	0.425 *
	Plot Twist	0.548 *	E6. Panel Number	Single Panel	0.960 ***
	Sarcasm	0.139 +		Multiple Panels	0.960 ***
	Wordplay Techniques	0.709 **	E7. Shot	Close Up	0.515 *
	- Homophonic Puns	1.000 ***		Medium Shot	0.541 *
	- Double Entendres	0.649 **		Long Shot	1.000 ***
	- Phonetic Mix-Ups	0.719 **	E8. Movement	Physical	0.508 *
	Realism	0.435 *		Emotion	0.461 *
E2. Subject	Human	0.869 ***		Causal	0.044 +
	Creature	0.946 ***		No Movement	0.508 *
	Cartoon Character	0.857 ***	E9. Textual Valence	Positive	0.566 *
	Object	0.568 *		Negative	0.597 *
	Scene	0.463 *		Neutral	0.436 *
E3. Cultural Specifics	Fictional Characters	0.762 **	E10. Emotion Types of Characters	Positive	0.790 **
	Public Figures	0.531 *		Negative	0.675 **
	Student Lifestyle	0.403 *		Neutral	0.522 *
	No Cultural Specifics	0.703 **		No Emotion	0.962 ***
E4. Stressor-Related or not	Related	0.716 **			
	Not Related	0.716 **			

which enables students to empathize and connect emotionally with the content [8], leading to more effective stress reduction.

- **Human Character:** “Human Character” was highlighted as the second important factor by the Logistic Regression model, presenting in 45.5% of effective memes but only 9.09% of ineffective ones. Memes depicting human characters are highly relatable to users. Seeing others who resemble themselves facing similar challenges fosters a sense of empathy and reduces feelings of isolation [10].
- **Simple Metaphor:** The Logistic Regression algorithm identified “Simple Metaphor” as a key feature, with 27.3% of effective memes utilizing it, compared to only 9.09% of ineffective memes. Metaphors in memes can contribute to the humorous framing of tough situations, therefore positively influencing emotions and relieving stress [15].
- **Positive Textual Valence and Positive Character Emotion:** The features “Positive Textual Valence” and “Positive Character Emotion” were identified as significant, with the former appearing in 27.3% of effective memes (compared to 9.09% of ineffective ones) and the latter present in 63.6% of effective memes (compared to 27.3% of ineffective ones). Positive textual messages in memes can encourage users and boost their self-confidence, which can reduce stress by diminishing feelings of helplessness. Also, characters displaying positive emotions in memes can trigger emotional contagion, prompting viewers to subconsciously mirror and internalize these emotions [11]. This effect can improve their mood and help alleviate exam-related stress.

Table 2: The overall performance of classifiers

Classifier	Accuracy $\pm$ SD	Precision	Recall	F1
KNN	0.528 $\pm$ 0.144	0.636	0.700	0.667
SVC	0.527 $\pm$ 0.120	0.583	0.700	0.636
Logistic Regression	0.592 $\pm$ 0.152	0.636	0.700	0.667
MLP	0.592 $\pm$ 0.086	0.500	0.600	0.545
Random Forest	0.481 $\pm$ 0.125	0.538	0.700	0.609

3 Discussion

First, across our features, elements that make the meme’s emotional meaning immediately legible (e.g., **close-up shots** and **salient facial expressions**) are associated with higher perceived stress-relief effectiveness. Analytically, this points to interpretability as a core dimension for exam-prep meme content. When viewers are cognitively loaded, memes that require fewer inferential steps to “get the feeling” may be more likely to be appraised as helpful. In line with stress appraisal accounts, clearer affective cues may reduce interpretive burden and support faster, less effortful appraisal of the situation [18]. This also suggests that, in stressful contexts, “readability” of affective meaning may be as important as the meme topic itself when designing or curating supportive content. For future HCI designs that incorporate memes as lightweight mental support, this suggests prioritizing compositions where the key affective cue is clear and not visually diluted by complex scenes or hard-to-read subjects.

Second, the association between **human subjects** and effectiveness labels suggests that perceived support may depend on whether

viewers can readily project themselves into the situation depicted. This may reflect stronger self-relevance and identification, which can shape stress appraisal and coping responses [18]. Moreover, human-centered depictions may also facilitate perceived social connection, a common coping resource in online contexts [2]. For future HCI designs that use memes in exam-preparation contexts, a practical direction is to depict exam-relevant experiences through characters that afford self-identification. This may be particularly appropriate when the goal is perceived stress support rather than pure entertainment.

Third, we find that **positive textual valence** and **positive character emotion** are more common among memes labeled effective. This may suggest that relatability alone is insufficient. Exam-stress memes that only amplify negativity can validate feelings but may leave viewers in the same affective state. In contrast, content that acknowledges the stressor while nudging the emotional direction upward may be more likely to be perceived as stress-relieving. From an emotion-regulation perspective, this pattern is consistent with reappraisal mechanisms that shift how a stressor is construed and experienced [5, 10, 13]. At the content level, this highlights valence as a controllable design dimension that may help differentiate “venting” memes from memes perceived as more supportive during exam preparation. For future HCI designs that incorporate memes in exam-preparation contexts, this points to valence as a deliberate design choice rather than an incidental byproduct of the template.

Taken together, these implications translate our element-level findings into practical guidance for meme content design and curation in exam-stress contexts. The resulting codebook and element-level associations can be reused as a structured content layer in future meme-based exam-stress support tools and meme-centered affective systems. They may also inform emerging GenAI meme-generation workflows by turning otherwise vague prompts into explicit, testable constraints on humor mechanism, subject depiction, and affective framing.

4 Conclusion

In this work, we curated a multi-platform dataset of 252 exam-related image memes, developed an annotation codebook via iterative content analysis to characterize meme design elements, and conducted a crowdsourcing study to capture students’ perceived stress-relief effectiveness of these memes in exam-preparation scenarios. We further linked the annotated design elements to perceived effectiveness through feature analysis to identify the most informative elements, and we discuss implications for meme content design and curation in exam-preparation contexts.

References

- [1] Umair Akram and Jennifer Drabble. 2022. Mental health memes: beneficial or aversive in relation to psychiatric symptoms? *Humanities and Social Sciences Communications* 9, 1 (Oct. 2022), 1–6. <https://doi.org/10.1057/s41599-022-01381-4> Publisher: Palgrave.
- [2] Umair Akram, Jennifer Drabble, Glhenda Cau, Frayer Hershaw, Ashileen Rajenthiran, Mollie Lowe, Carissa Trommelen, and Jason G. Ellis. 2020. Exploratory study on the role of emotion regulation in perceived valence, humour, and beneficial use of depressive internet memes in depression. *Scientific Reports* 10, 1 (Jan. 2020), 899.
- [3] Kristine Ask and Crystal Abidin. 2018. My life is a mess: Self-deprecating relatability and collective identities in the memification of student issues. *Information, Communication & Society* 21, 6 (2018), 834–850.
- [4] Amit Baume, Theresa Fleming, and Stephen M. Schueller. 2020. Digital Micro Interventions for Behavioral and Mental Health Gains: Core Components and Conceptualization of Digital Micro Intervention Care. *Journal of Medical Internet Research* 22, 10 (Oct. 2020), e20631. <https://doi.org/10.2196/20631>
- [5] Shannon T Brady, Bridgette Martin Hard, and James J Gross. 2018. Reappraising test anxiety increases academic performance of first-year college students. *Journal of educational psychology* 110, 3 (2018), 395.
- [6] Marie Louise Caltabiano. 1994. Measuring the similarity among leisure activities based on a perceived stress-reduction benefit. *Leisure Studies* 13, 1 (1994), 17–31.
- [7] Patrick Davison. 2020. 9. The Language of Internet Memes. In *The Social Media Reader*, Michael Mandiberg (Ed.). New York University Press, 120–134. <https://doi.org/10.18574/nyu/9780814763025.003.0013>
- [8] Paul Ekman and Wallace V Friesen. 2003. *Unmasking the face: A guide to recognizing emotions from facial clues*. Vol. 10. Ishk.
- [9] Tuncay Ergene. 2003. Effective interventions on test anxiety reduction: A meta-analysis. *School psychology international* 24, 3 (2003), 313–328.
- [10] James J Gross. 2002. Emotion regulation: Affective, cognitive, and social consequences. *Psychophysiology* 39, 3 (2002), 281–291.
- [11] Elaine Hatfield, John T Cacioppo, and Richard L Rapson. 1993. Emotional contagion. *Current directions in psychological science* 2, 3 (1993), 96–100.
- [12] Eva M Hernández-Cuevas and Nelson D Cruz-Bermudez. 2021. Meme sharing “culture” and psychological well-being from a neuroscientific framework. *Revista [IN] Genios* 7, 2 (2021), 1–20.
- [13] Jeremy P Jamieson, Wendy Berry Mendes, Erin Blackstock, and Toni Schmader. 2010. Turning the knots in your stomach into bows: Reappraising arousal improves performance on the GRE. *Journal of experimental social psychology* 46, 1 (2010), 208–212.
- [14] Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. 2020. The hateful memes challenge: Detecting hate speech in multimodal memes. *Advances in neural information processing systems* 33 (2020), 2611–2624.
- [15] Zoltan Kovecses. 2010. *Metaphor: A practical introduction*. Oxford university press.
- [16] Peter J Lang, Margaret M Bradley, Bruce N Cuthbert, et al. 1997. International affective picture system (IAPS): Technical manual and affective ratings. *NIMH Center for the Study of Emotion and Attention* 1, 39–58 (1997), 3.
- [17] Emily G. Lattie, Rachel Kornfield, Kathryn E. Ringland, Renwen Zhang, Nathan Winquist, and Madhu Reddy. 2020. Designing Mental Health Technologies that Support the Social Ecosystem of College Students. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–15.
- [18] Richard S Lazarus. 1984. *Stress, appraisal, and coping*. Vol. 464. Springer.
- [19] Chen Ling, Ihab AbuHilal, Jeremy Blackburn, Emiliano De Cristofaro, Savvas Zannettou, and Gianluca Stringhini. 2021. Dissecting the Meme Magic: Understanding Indicators of Virality in Image Memes. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (April 2021), 1–24.
- [20] Rod A Martin and Thomas Ford. 2018. *The psychology of humor: An integrative approach*. Academic press.
- [21] Philipp Mayring. 2014. Qualitative content analysis: theoretical foundation, basic procedures and software solution. (2014).
- [22] Mary L McHugh. 2012. Interrater reliability: the kappa statistic. *Biochemia medica* 22, 3 (2012), 276–282.
- [23] Ryan M. Milner. 2016. *The world made meme: public conversations and participatory media*. The MIT Press, Cambridge, Massachusetts.
- [24] Jessica Gall Myrick, Robin L. Nabi, and Nicholas J. Eng. 2022. Consuming memes during the COVID pandemic: Effects of memes and meme type on COVID-related stress and coping efficacy. *Psychology of Popular Media* 11, 3 (July 2022), 316–323.
- [25] Mohi Reza, Angela Zavaleta Bernuy, Emmy Liu, Tong Li, Zhongyuan Liang, Calista K Barber, and Joseph Jay Williams. 2023. Exam eustress: Designing brief online interventions for helping students identify positive aspects of stress. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [26] David Robotham and Claire Julian. 2006. Stress and the higher education student: a critical review of the literature. *Journal of Further and Higher Education* 30, 2 (May 2006), 107–117.
- [27] Iris K Schneider, Lotte Veenstra, Frenk van Harreveld, Norbert Schwarz, and Sander L Koole. 2016. Let’s not be indifferent about neutrality: Neutral ratings in the International Affective Picture System (IAPS) mask mixed affective responses. *Emotion* 16, 4 (2016), 426.
- [28] Limor Shifman. 2013. *Memes in digital culture*. MIT press.
- [29] Viriya Taecharungroj and Pitchanut Nueangiamnong. 2015. Humour 2.0: Styles and Types of Humour and Virality of Memes on Facebook. *Journal of Creative Communications* 10, 3 (Nov. 2015), 288–302. <https://doi.org/10.1177/0973258615614420>
- [30] Moshe Zeidner. 1998. *Test anxiety: the state of the art*. Plenum Press, New York.